

CryptoChain Analyzer Dashboard

1. Cryptographic Metrics

1.1 Proof of Work and SHA-256 (M1, M2)

Bitcoin's security is based on SHA-256, a cryptographic hash function that produces a fixed 256-bit output. The Proof of Work (PoW) mechanism requires miners to find a nonce such that the double hash of the block header $\text{SHA256}(\text{SHA256}(\text{header}))$ is below a target threshold. This target is encoded in the compact **bits** field of every block header using the formula:

$$\text{target} = \text{coefficient} \times 2^{(8 \times (\text{exponent} - 3))}$$

The dashboard verifies this locally using Python's **hashlib** library: the raw 80-byte header is fetched from the API, double-hashed, and compared against the computed target. The number of leading zero bits in the resulting hash indicates the difficulty. Each additional zero bit doubles the expected mining effort. Current blocks show approximately 90 or more leading zero bits, confirming the enormous computational effort required to produce a valid block.

1.2 Difficulty Adjustment (M3)

Every 2,016 blocks (approximately two weeks), the Bitcoin protocol adjusts the mining difficulty to keep the average inter-block time at 600 seconds. The adjustment formula described in the Bitcoin whitepaper is:

$$\text{new_difficulty} = \text{old_difficulty} \times (\text{actual_time} / 1,209,600\text{s})$$

The dashboard plots the difficulty over the last adjustment periods and displays the ratio of actual block time to the 600-second target for each period. Ratios above 1.0 indicate miners were faster than expected (difficulty increases); ratios below 1.0 indicate they were slower (difficulty decreases).

1.3 Merkle Trees (M5)

Each Bitcoin block commits to all its transactions via a Merkle tree, a binary tree where each leaf is a transaction ID (txid) and each internal node is $\text{SHA256}(\text{SHA256}(\text{left} + \text{right}))$. The root of this tree is stored in the block header as the **Merkle Root**. The dashboard builds the full Merkle tree for any selected block and generates the proof path for a chosen transaction, verifying step by step that the computed root matches the header. This demonstrates the logarithmic efficiency of Merkle proofs: only $\log_2(n)$ hashes are needed to verify any transaction in a block of n transactions.

1.4 Network Hash Rate and 51% Attack Security (M1, M6)

The estimated network hash rate is derived from the current difficulty:

$$\text{hashrate} = \text{difficulty} \times 2^{32} / 600$$

This figure, currently in the range of several hundred exahashes per second (EH/s), represents the total computational power securing the network. Module M6 uses this value to estimate the cost of a hypothetical 51% attack, the amount of hardware and electricity required to control the majority of hash rate. The probability of a successful attack as a function of confirmation depth is modelled using the

Poisson approximation from Nakamoto (2008), Section 11, showing that 6 confirmations provide near-complete security against attackers with up to 30% of network hash rate.

2. AI Components

2.1 Anomaly Detector — M4

Model choice: Because each hash attempt is independent, the time between consecutive Bitcoin blocks follows an exponential distribution with mean 600 seconds, a direct consequence of the memoryless property of the Poisson process. Significant deviations from this distribution may indicate unusual mining pool behaviour, network partitions, or statistical outliers.

Method: Two independent statistical tests are applied to each inter-block time. First, a Z-score is computed on the log-transformed times (the log transform makes the distribution more symmetric). Second, the exponential CDF is used to compute a p-value, the probability of observing a value as extreme as the measured time under the null hypothesis. A block is flagged as anomalous only if both tests agree ($|Z| > 2.0$ and $p\text{-value} < 0.05$), reducing false positives.

Evaluation: The model is compared against a simple threshold baseline (flag if inter-block time > 20 min or < 1 min) using Precision, Recall, and F1. Ground truth is defined as blocks flagged by the p-value test alone. Using two tests jointly typically improves Precision at the cost of Recall compared to the baseline.

Model	Precision	Recall	F1
Z-score + p-value (our model)	Higher	Lower	Comparable
Simple threshold (baseline)	Lower	Higher	Comparable

Table 1. Qualitative comparison of anomaly detection models.

2.2 Difficulty Predictor — M7

Model choice: Bitcoin's mining difficulty has shown a consistent long-term upward trend as more miners join the network over time. A Linear Regression model on time-indexed difficulty values is a natural first approach: interpretable, fast to train, and evaluable with standard regression metrics. Unlike LSTM or Prophet, it has no hidden components and is easy to explain.

Training: The model is trained on the last 20 difficulty adjustment points fetched from the blockchain, using the period index (1, 2, 3, ...) as the single feature and the difficulty value (in Trillions) as the target. The last 3 periods are held out as a test set.

Evaluation: The model is evaluated on the held-out test set using MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R2. Results indicate that the model captures the long-term trend well ($R^2 > 0.5$ in stable periods) but underestimates sharp drops, as observed in early 2026 when difficulty fell from ~155T to ~125T. This is a known limitation of linear models applied to non-monotonic time series.

Comparison with M4: The two AI models are complementary. M4 monitors real-time block-level anomalies using a distribution-based statistical test; M7 provides longer-term forecasts of network difficulty using a supervised regression model. They use different data, different methods, and different evaluation metrics.

	M4 — Anomaly Detector	M7 — Difficulty Predictor
Problem	Detect abnormal block times	Forecast next difficulty value

Model	Z-score + Exponential CDF	Linear Regression
Data	Last ~25 inter-block times	Last ~20 difficulty adjustments
Evaluation	Precision, Recall, F1	MAE, RMSE, R2
Limitation	Small sample size (~25 blocks)	Linear model may underfit sharp changes

Table 2. Comparison between M4 and M7 AI models.

3. References

- [1] S. Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System, 2008. <https://bitcoin.org/bitcoin.pdf>
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